(Please write your Enrollment Number) Enrollment No. 04801172022

MID-TERM EXAMINATION

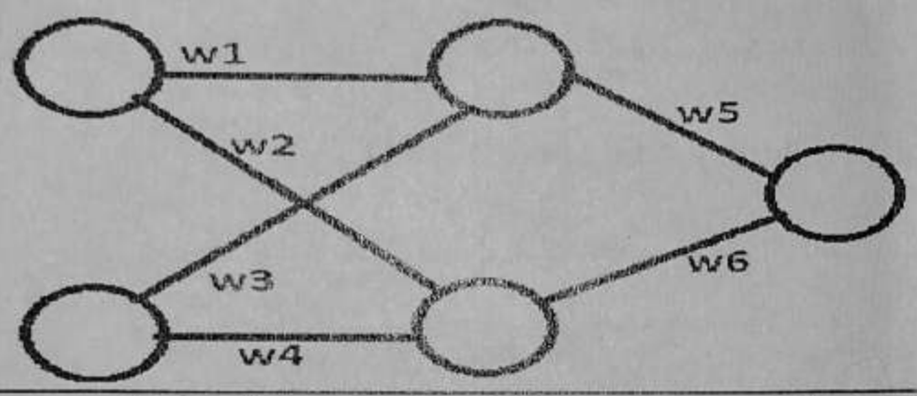
(Course Name : B. Tech.) (Semester : 5th)

(October 2024) OFF LINE mode

Subject Code: BAI-305	Subject: Deep Learning-I
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a) Identify the type of problems solvable by MP-Neuron?		CO1
	b) Explain the perceptron learning algorithm.		CO1
	c) In an ANN, which process is followed to obtain the network output? Explain with example.		CO2
	d) Identify whether $f(x) = x^4 - 2x^3 + 2$ is suitable for Gradient Descent? Justify with suitable reason.		CO2
	UNIT I		
Q2	Attempt any two parts	(5*2=10)	CO Mapping
	a) In a two-input sigmoid neuron, each edge with weights (w1, w2) are initialized as (0.1,-0.2) and bias as 0.1 with cost function as $(y^2-y^*)^2$, where 'y' is the target output and 'y*' is the estimated output. On applying gradient descent algorithm with learning rate as 0.2 on this neuron, what are the values of (w1, w2) after two iterations on a dataset with two data samples as (1,2) and (0,5) with output labels as 0 and 1, respectively.		CO1
	b) Why can't we use a single equation to update the weights for positive and negative samples in the perceptron learning algorithm? Justify with proof.		CO1
	c) 'Perceptron is suitable for linear problem only'. Justify this statement with proof.		CO1
	UNIT II		
Q3	Attempt any two parts	(5*2=10)	CO Mapping
	a) Is a neural network comprising a single hidden layer capable of effectively representing and solving all possible Boolean functions where number of Boolean inputs are quite high? Justify your answer with appropriate reason		CO2
	b) Consider the following artificial neural network, each neuron in the hidden layer and output layer has a bias of 0.1. The activation function used is the sigmoid function, and the cost function is the squared error		CO2

	function. The input sample values are (3, 0.6), and the output label is 0.5. The weights are initialized for two scenarios: (i) (w1,w2,w3,w4,w5,w6) = (0.1,0.5,1,1,0.2,0.5) (ii) (w1,w2,w3,w4,w5,w6) = (0.2, 1, 0.5, 1, 0.1, 0.1). According to gradient descent algorithm, what are the updated weights after performing a single backpropagation step for the weight initialization scenarios (i) and (ii)? 		
	c) According to gradient descent algorithm, compute the weight updation equation for w1 and w4 for the ANN (illustrated in Q3. (b)), where cost function as squared error function and activation function as tanh(x) whose derivative is $(1-(\tanh(x))^2)$.		CO2